

When Safe Search Budgets Do Not Transfer Across Graph-ANNS Rebuilds: [Experiments & Analysis]

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Abstract

Rebuilding a graph index can change the search procedure on which an approximate nearest-neighbor search budget was calibrated. We study when a source-build budget retains its recall target on a rebuilt index, how this transfer should be audited, and what target-side recovery costs. ICBA organizes the analysis around build-conditioned risk, isolated selection and certification roles, and explicit search-cost accounting. In a SIFT-100K and Arxiv-Nomic-100K query-profile replay after 5% data refresh and rebuilding, direct TCP pool reuse has empirical target failure rates of 6.69% and 7.24%. Target-selection recalibration reduces these rates to 1.79% and 2.04%, with mean distance-count reductions of 44.39% and 44.79% relative to the registered fixed endpoint. Cross-family and scale analyses extend the evidence beyond this recovery case. The central systems implication is to treat a rebuilt index as a new calibration environment: measure portability, separate observed recovery from a valid certificate, and account for the cost of obtaining target information.

Keywords

approximate nearest-neighbor search, graph indexes, index rebuilding, risk auditing, budget transfer

W2 writing draft. Introduction and Related Work are drafted with verified references; later sections remain a structural draft. Numeric exhibits use the W0 frozen macros. Outlined exhibits are planned placements, not experimental plots. Section 9.1 lists the remaining writing and evidence checks.

1 Introduction

Search quality depends on both the graph index and the budget used to explore it. The index determines which paths a query can traverse, while the budget controls how much of that search is explored. HNSW and Vamana make this tradeoff central to efficient approximate nearest-neighbor search [9, 11]. After rebuilding, a budget selected on the previous graph may execute on a different search structure. Preserving the parameter value does not establish that its recall behavior is preserved. This creates an operational question at the boundary between index maintenance and query serving: *when can an existing budget policy be reused, and when is target-side information worth acquiring?*

The question has two distinct settings. Rebuilding an unchanged vector collection can alter the graph through construction order or randomization. Rebuilding after a data refresh additionally changes the collection and potentially the nearest-neighbor ground truth. Prior work documents construction sensitivity and develops efficient update mechanisms [6, 10]. Adaptive search methods address another part of the problem: allocating work to individual queries using predicted recall or search cost [4, 8, 12]. These capabilities make budget reuse attractive, but its value

still depends on the relationship between the source calibration environment and the target index.

Evaluating that relationship requires more than comparing average recall. A policy can attain a high mean while a subset of queries falls below the requested quality. We study the probability that a query’s recall is below a specified threshold, separately from the expected fraction of missed neighbors. In the primary experiment, failure means $\text{Recall@10} < 0.95$: because recall is discrete, even one missing neighbor is a failure. A 5% risk target therefore asks that at least 95% of queries recover all ten neighbors. This event is stricter than mean recall of 0.95 and must remain explicit when interpreting external methods.

Our organizing insight is that budget portability belongs to a *source policy, target build, and query workload*. ICBA turns this dependence into an audit. It records what information determines each action, separates target selection from certification and evaluation, and distinguishes absolute target failure from additional failure relative to a reference. The theoretical analysis is organized around the observation assumptions under which source information can or cannot support a target decision. The empirical analysis asks how often reuse fails, which evidence supports recovery, and what that recovery costs. TCP provides the recovery case study: it reuses historical query-budget profiles and adjusts their decisions with target-selection information, followed by a certification decision and fallback.

The primary replay covers SIFT-100K and Arxiv-Nomic-100K, with ten target builds per dataset after a 5% mixed delete/insert refresh. Direct TCP pool reuse yields empirical target failure rates of 6.69% and 7.24%. Target-selection recalibration reduces them to 1.79% and 2.04%, while reducing mean distance calculations by 44.39% and 44.79% relative to the registered fixed endpoint. These results concern queries with corresponding old-build profiles; they measure empirical recovery in that replay population. Search savings and a valid certificate for the entire candidate/fallback decision are separate outcomes.

The broader evidence is deliberately layered. External-method bridges examine query-level risk under their supported interfaces, while Vamana and Deep1M analyses examine different graph-family and scale conditions. These blocks retain their own references and failure definitions. Recovery is then evaluated against its acquisition cost: obtaining target labels and profiles can outweigh cheaper searches over a short serving horizon. Thus, an operator needs both a risk assessment and an amortization calculation before replacing direct reuse with recalibration.

We make four contributions:

- (1) A build-conditioned formulation that separates absolute risk, reference-relative transfer loss, and the information available to the budget policy.

Planned exhibit: Source-to-target budget transfer

Show the same query workload passing through source calibration and a rebuilt target graph. Place three decisions at the target: reuse, recalibrate, or fixed endpoint. Mark the location of risk measurement and information-acquisition cost.

Figure 1: Overview placement. Scientific question: which part of a calibrated search decision changes when the graph is rebuilt? This schematic will explain the setting rather than depict an unmeasured effect.

- (2) An ICBA audit workflow connecting query roles, discrete failure events, certification decisions, and search-cost accounting.
- (3) A layered empirical study of budget portability across the registered recovery, method-bridge, graph-family, and scale settings.
- (4) A TCP recovery case study that quantifies the search benefit of target information and its workload-dependent acquisition cost.

Together, these contributions support treating the search policy as part of the index’s versioned operational state, with an explicit decision about reuse after rebuilding.

2 Related Work and Positioning

Graph construction, evaluation, and maintenance. HNSW combines hierarchical navigation with graph search, while DiskANN introduces Vamana as part of a system for large-scale search [9, 11]. Azizi et al. [3] compare graph-search methods and construction paradigms across datasets and scales. Elliott and Clark [6] show that insertion order and intrinsic dimensionality affect HNSW recall. These studies establish that the constructed graph matters to search quality. We examine a different experimental unit: a budget policy transferred from a source build to a target build, with a matched reference used to separate target difficulty from transfer loss. FreshDiskANN [10] supports insertions and deletions without repeatedly reconstructing the entire index. Such maintenance mechanisms motivate distinguishing graph-only rebuilding from refresh-plus-rebuild; the latter is the setting of our primary TCP replay.

Adaptive work allocation and declarative recall. LAET predicts query-specific termination conditions using both static and intermediate-search features [8]. DARTH instead predicts achieved recall during execution and adapts prediction intervals to terminate HNSW or IVF searches at a declarative target [4]. ANNiE estimates the work required to reach a recall target from graph-derived features; its ANNiE-S search uses those estimates directly [12]. Ada-ef models query-to-dataset similarity distributions to choose an HNSW exploration parameter [13]. It also studies incremental updates to its statistics and estimation table, including comparisons with stale and fully recomputed variants after insertions or deletions. Our question concerns the validity and cost of carrying a decision between builds. It applies to adaptive policies as well as fixed budgets, but comparisons must retain each policy’s permitted signals and update procedure. The frozen

DARTH and Ada-ef bridges are evidence about those integrations, rather than a common ranking of all published methods.

Quality guarantees and their target quantities. ConANN applies conformal risk control to IVF search, controlling expected false-negative rate while adapting the number of probed clusters [7]. Its objective is the expected fraction of missed neighbors, whereas our principal event is whether a query falls below a recall threshold. ANNiE’s optional recall-target guarantee directly addresses the fraction of queries reaching the requested recall [12]. Its analysis uses quantile-loss minimization, and its evaluation examines both guarantee deviations and out-of-distribution queries. The DARTH+ preprint also introduces optional probabilistic quality guarantees for HNSW and IVF [5]. Consequently, the distinction of our study is not the existence of a quality guarantee. We focus on which policy and build a claim covers, what information is reused or reacquired, and how the resulting decision behaves after rebuilding. A failure rate computed with our event does not, by itself, refute a guarantee for a different loss.

Calibration, selection, and the role of ICBA. Conformal risk control bounds an expected monotone loss under its calibration assumptions [2]. Learn then Test formulates risk-controlled parameter selection as multiple hypothesis testing [1]. These are statistical foundations for auditing a search decision, rather than guarantees that automatically survive a changed execution environment. ConANN explicitly aligns calibration with index construction and assumes exchangeable query data [7]. In our setting, rebuilding can change the loss induced by the search even when the query population is retained. ICBA makes the target build, failure event, candidate selection, and fallback rule explicit. TCP supplies a concrete query-profile reuse and recovery case within that audit. This separation lets us ask whether target information restores useful search efficiency without conflating empirical recovery, a pipeline-level certificate, and net lifecycle savings.

3 Problem Formulation and Theoretical Lens

Section purpose: Define the estimands first, then state exactly what a theory result would explain.

Let G denote an index build, $q \sim \mathcal{Q}$ a query, and $a \in \mathcal{A}$ a registered search action. A policy π maps its permitted observations to an action. For recall target τ , define

$$\begin{aligned} Z_G(q, a) &= \mathbf{1}\{\text{Recall}@k(G, q, a) < \tau\}, \\ \mathcal{R}_G(\pi) &= \mathbb{E}_{q \sim \mathcal{Q}} Z_G(q, \pi(q)). \end{aligned} \quad (1)$$

The observation arguments of π are suppressed in the notation, not in the experimental contract. In the primary replay $k = 10$ and $\tau = 0.95$; with discrete recall this threshold requires ten of ten neighbors to be retrieved. It is a stricter event than failure to retrieve nine of ten.

Absolute and incremental risk. For a policy obtained on source build G_s and a specified target reference π_t^{ref} , use

$$\Delta_{s \rightarrow t} = \mathcal{R}_{G_t}(\pi_s) - \mathcal{R}_{G_t}(\pi_t^{\text{ref}}). \quad (2)$$

The absolute target risk and the reference risk must accompany this difference. A difference expressed in percentage points is not

a relative percentage change. The reference and query distribution are part of the estimand.

Budget actions, work, and censoring. Requested search parameters, distance evaluations, and wall-clock time are separate quantities. A larger requested action need not imply a monotone raw recall sequence or a fixed amount of work. When an experiment defines failure using an under-budget or endpoint-censored event, it requires its own event definition and cannot inherit Equation (1) by relabeling. Endpoint truncation will be shown together with any minimum-budget summary.

Conditional information limits. The theoretical slot is a conditional impossibility result: if the observations available to a decision rule do not distinguish rebuild states that require incompatible actions, uniformly safe and efficient transfer may be impossible. The final statement must specify the observed variables, admissible policies, loss, state distribution, and the hypothesis controlling distinguishability. The old empirical total-variation premise is not used here as an established measurement. Proof and empirical-premise reconciliation remain part of the next writing stage.

Certification of one fixed policy. Let x failures be observed among n independent certification queries after the policy has been fixed independently of those queries. A one-sided Clopper-Pearson upper confidence bound is

$$U(x, n; \alpha) = \begin{cases} \text{Beta}^{-1}(1 - \alpha; x + 1, n - x), & x < n, \\ 1, & x = n. \end{cases} \quad (3)$$

Here α controls the coverage error and δ is the tolerated failure probability. The acceptance rule is $U \leq \delta$. Even when both are set to 0.05, their meanings differ. Applying this bound to several actions and choosing one afterward requires an additional joint-validity argument. It does not follow directly from the single-policy bound.

4 ICBA Workflow and the TCP Recovery Case

Section purpose: Make the measurement and decision path executable in the reader’s mind.

ICBA records the graph build, allowed observations, candidate action family, query roles, risk event, and cost units before interpreting a recovery result. Selection, certification, and evaluation have distinct functions. Selection chooses an action using the permitted target information. Certification evaluates a frozen candidate under a stated statistical rule. Evaluation estimates its behavior after selection and certification are complete.

Query-profile source reuse. In the frozen TCP replay, each query has a profile of minimum passing actions from nine old builds, excluding the target-seed build. The source-pool action is the maximum of these old-build minima. This is a query-indexed profile-reuse workload: the old-build responses for the corresponding query are inputs to the policy. A workload of previously unprofiled queries would require an additional acquisition or prediction path.

Target recalibration. The target-selection procedure chooses a global shift on the registered action grid. It then checks the selected candidate on certification queries. The implementation also

Planned exhibit: Audit and recovery path

Inputs: graph identity, source query profiles, action grid. Separate target selection, certification, and evaluation lanes. Route a rejected candidate to the registered endpoint. Attach profile, truth, certification, search, and fallback costs to the operations that incur them.

Figure 2: Workflow placement. Scientific question: what information and costs produce each deployment decision? Certification composition remains a separate proof obligation.

checks the endpoint on that certification role and may choose it as fallback. We retain the resulting empirical behavior in the recovery tables. A pipeline-wide confidence statement depends on resolving the candidate/endpoint selection composition, rather than interpreting each individual upper bound as a joint guarantee.

Fallback is an action. The registered fixed endpoint is $ef = 200$ in the primary case. Its failure event and cost are measured like those of other actions. Calling it an endpoint identifies its place on the grid; it does not establish zero risk. Report both the fallback-build count and its contribution to mean and tail search work.

5 Experimental Methodology

Section purpose: Give each evidence block its own population, unit, and event definition.

Primary recovery block. The current numerical tables use the frozen SIFT-100K and Arxiv-Nomic-100K query-profile replay after a 5% mixed delete/insert refresh and rebuilding, with ten target builds per dataset. This block combines data change with graph change; it does not isolate the effect of rebuilding an unchanged collection. It uses 500 target-selection, 500 certification, and 1,000 evaluation queries. The complete setup table will identify source-build history, query IDs and overlap checks, vector snapshot, grid, implementation commit, and hardware for each block. These fields must come from the corresponding manifests before a unified setup is asserted.

Metrics. We report empirical failure rates, mean distance-count gain, sample query-pooled p95 distance counts, and fallback-build counts. Let $\bar{c}_{\text{end}}$ and $\bar{c}_{\pi}$ denote the mean work under the registered endpoint and a policy. Search gain is

$$g(\pi) = 1 - \frac{\bar{c}_{\pi}}{\bar{c}_{\text{end}}}. \quad (4)$$

Negative values mean additional work. A lower sample p95 is a descriptive tail comparison; it is not a statistical noninferiority certificate or a latency measurement.

Uncertainty and dependence. Build-level variation is distinct from query variation. The current gain intervals use the frozen nested-bootstrap implementation, which independently resamples queries within each sampled build. Since query IDs may recur across builds, the dependence structure and intended estimand require reconciliation before assigning a population-level interpretation to these intervals. The final robustness exhibit will report

Table 1: Planned setup matrix: fields that must be traceable for each evidence block. This is a reporting schema, not a completed experimental inventory.

Block	Required setup fields
Primary replay	Source/target builds; historical query profiles; grid; role counts; risk event
Method bridges	Published method; adapter changes; permitted observations; matched threshold
Cross-family	Native action semantics; reference policy; censoring; source qualification
Scale	Dataset size; build pairs; query overlap; reference and incremental risks
Economics	Cost unit; shared/sunk inputs; acquisition policy; serving horizon

Table 2: Frozen primary replay results. Risk is empirical Recall @10 < 0.95 failure; gain is mean distance-count reduction against the registered endpoint. These rows use query-indexed old-build profiles.

Dataset	Policy	Risk (%)	Gain (%)
SIFT-100K	Source pool	6.69	59.51
	Target shift	1.79	44.39
Arxiv-100K	Source pool	7.24	64.21
	Target shift	2.04	44.79

build-wise outcomes, the registered leave-one-build-out analysis, and the uncertainty procedure actually used.

Cross-block comparisons. External-method bridges, Vamana reanalysis, and Deep1M use separate protocols. Their absolute risks are not pooled into a common method ranking. A comparison becomes quantitative across blocks only after matching recall event, source qualification, query population, reference action, and information budget.

6 Budget Portability on Rebuilt Graphs

Section purpose: Establish the phenomenon before discussing recovery.

Table 2 shows the primary contrast. Direct source TCP pool reuse has empirical target failure of 6.69% on SIFT and 7.24% on Arxiv. Both exceed the 5% target, despite reducing mean distance counts relative to the fixed endpoint. Thus, in this replay, low search work and target recall compliance are different outcomes.

The target-selection lane reduces the failure rates to 1.79% and 2.04%. Its mean search savings remain 44.39% and 44.79%. This contrast motivates the recovery analysis: target information changes the operating point, but its value must include acquisition costs and the validity of the resulting decision rule.

Mechanism analysis to attach. The source-to-target matrix should show which builds and queries drive failure. Separate index changes, source-policy qualification, target difficulty, and endpoint censoring before attributing a gap to a single cause. A deterministic-contract or margin-based explanation will enter this

Planned exhibit: Transfer matrix and risk-work trade-off

Panel A: source-to-target risk matrix with reference cells. Panel B: per-build risk versus mean distance counts for source reuse, target recalibration, and endpoint. Include the 5% risk target without treating it as a confidence bound.

Figure 3: Phenomenon placement. Scientific question: is transfer failure systematic or concentrated in a small subset of builds? Per-build frozen inputs are required before plotting.

Table 3: Evidence map across protocols. Values are scoped observations, not a common leaderboard. The event and reference for each row must accompany the expanded experiment description.

Evidence block	Frozen result and interpretation
DARTH bridge	SIFT 22.55%; Arxiv 23.12% target failure in the registered adapter
AdaEF bridge	Arxiv 9.52% target failure in its bridge protocol
Vamana audit	SIFT 16.86%; Arxiv 11.37%; post-hoc cross-family evidence
Deep1M	20.04 pp increment over 8.82% reference risk

section only after its event and evidence are aligned with the primary results.

7 Cross-Method, Cross-Family, and Scale Evidence

Section purpose: Extend the phenomenon while keeping protocol differences visible.

Registered method bridges. The DARTH bridge records 22.55% and 23.12% target failure on SIFT and Arxiv, respectively. The AdaEF bridge on Arxiv records 9.52%. These observations belong to the specific registered adapters and threshold settings. The final comparison will state what was retained from the original method and what changed for the transfer test.

Vamana. The aligned post-hoc audit reports 16.86% and 11.37% risk for SIFT and Arxiv, using six source builds, six distinct target builds, 36 source-target pairs, and 750 queries. This evidence adds a graph-family case. Its post-hoc status and source-qualification rule remain visible rather than being collapsed into the primary replay table.

Deep1M. The scale block contains 8 builds, 1000 queries, and 56 directed pairs. Transport risk is 28.86%, while the reference risk is 8.82%. Their frozen difference is 20.04 percentage points, with reported interval [17.29, 22.77]. The reference risk is already above 5%; the difference measures an additional transfer penalty, not failure of an independently established safe reference.

8 Recovery Quality and Lifecycle Economics

Section purpose: Turn the recovery example into an operational trade-off, not a universal algorithm ranking.

Planned exhibit: Layered evidence and boundary cases

Use separate panels for external-method bridges, Vamana, and Deep1M. Display the local event, target/reference pair, and uncertainty unit in each panel. Include negative and nontransfer cases when their frozen provenance is reconciled.

Figure 4: Scope placement. Scientific question: which parts of the phenomenon recur across settings, and which depend on the action class or source/reference definition?

Table 4: Target-selection replay: query-pooled sample p95 distance counts and fallback builds. Each dataset has ten target builds.

Dataset	Target p95	Endpoint p95	Fallback
SIFT	2349.05	2615.00	0/10
Arxiv	2974.10	3252.05	1/10

Search and tail behavior. For the target-selection lane, the frozen mean-gain intervals are [43.88, 44.91]% on SIFT and [34.72, 50.15]% on Arxiv. Their bootstrap interpretation follows Section 5. Table 4 provides the descriptive tail comparison and fallback counts. The lower observed p95 values refer to distance evaluations; they do not measure GPU/CPU latency or establish statistical tail noninferiority.

A common cost unit. In a distance-count model, let C_{acquire} include the nonshared profile, truth, selection, and certification work, and let $\bar{c}$ denote serving work per execution. For a fixed expected workload,

$$C_{\pi}(N) = C_{\text{acquire},\pi} + N\bar{c}_{\pi}, \quad (5)$$

$$N_{\text{BE}} = \frac{C_{\text{acquire},\pi} - C_{\text{acquire},\text{ref}}}{\bar{c}_{\text{ref}} - \bar{c}_{\pi}}, \quad (6)$$

when the denominator is positive and the acquisition difference is positive. Nonpositive acquisition difference means immediate modeled amortization if serving is cheaper; nonpositive serving savings require a separate case rather than a positive break-even claim. Shared costs cancel only when the competing strategies genuinely incur the same cost.

Conditional model outputs. The current dataset summaries report break-even at 138836 query executions for SIFT and 107378 for Arxiv. Zero SIFT targets and 1 Arxiv target have no positive modeled serving savings. These values condition on the source-profile cost treatment and reuse assumptions in the frozen model. Target truth is charged, but old-build truth is not explicitly included as a separate acquisition term. A valid lifecycle interpretation therefore requires its shared or sunk status to be specified symmetrically for the reference strategy.

Operational conclusion to test against the final evidence. TCP recovery is useful when the needed query profiles are available, target recalibration yields an acceptable operating point, and the serving horizon amortizes acquisition work. For a new-query

Planned exhibit: Recovery and amortization

Panel A: build-wise risk, mean NDC, and p95 for the registered actions. Panel B: modeled net work versus query executions, with explicit source-profile accounting scenarios and nonamortizing targets retained.

Figure 5: Recovery placement. Scientific question: when does an empirically favorable target policy repay the information needed to obtain it? Model curves must be labeled separately from measured search counts.

workload, the missing profile-acquisition path changes both deployment and economics. The final text will give these regimes rather than claim that one recovery strategy dominates at every horizon.

9 Reproducibility and Interpretation

Section purpose: State the consequences of the evidence boundaries once, at the level where they change the conclusion.

Evidence lineage. The numerical exhibits are generated from a frozen ledger linked to committed summary tables, per-build records, analysis code, and decision manifests. This writing stage does not regenerate raw query responses. The final artifact description will distinguish numerical consistency checks, replay reproducibility, and independent statistical validation.

Deployment population. The primary TCP result concerns queries with corresponding old-build profiles. Target-role separation does not make those source-side query profiles free or available for an unseen workload. This population definition is necessary both for a deployment claim and for amortization.

Theory and confidence. The theoretical component must state its observation model and prove its premises at the claimed level. An empirical classifier or a plug-in discrepancy does not automatically provide the upper bound required by an information-theoretic result. Likewise, separate single-action confidence bounds do not automatically certify a data-dependent candidate/fallback decision.

Implication for index operations. Budget policies should be versioned with the build and workload on which they were assessed. A rebuild can trigger a targeted audit without requiring a universal prescription to retrain every policy. The operator needs three separate answers: the new risk, the cost of a valid decision, and the horizon over which that decision is useful.

9.1 Writing Completion Map

This subsection is an internal writing aid and will be removed from the submission manuscript. Introduction and Related Work are drafted. Subsequent stages complete the setup matrix, resolve the theory premises and candidate/fallback certificate composition, check the shared-query resampling estimand, and make the source-profile acquisition assumptions explicit. Existing Faiss, deterministic-contract, and larger build-farm results require ledger extension before their headline values enter this draft.

The main paper budget is 12 pages excluding references. Full proofs and essential supplementary details are planned for a separate short appendix PDF. The present skeleton deliberately leaves space for verified exhibits rather than filling the budget with provisional claims.

10 Conclusion

Search-budget reuse across graph rebuilds is a measurable transfer decision. The frozen evidence shows target failures under direct source reuse and empirical recovery from target recalibration in the query-profile setting. ICBA connects those observations to explicit risk events, independent query roles, and lifecycle accounting. The resulting lesson is to evaluate portability and recovery together: retained search efficiency, statistical assurance, and the cost of acquiring target information describe different parts of the same operational decision.

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